05-03 第1篇

阅读理解（说明文） Why Do Onions Make Us Cry?

① Many students help their parents cook at home, and sooner or later they meet a small but surprising enemy in the kitchen — the **onion**. You cut it, and within seconds your eyes start to burn, your nose runs, and tears run down your cheeks. Nobody is sad. Nobody is hurt. So why does this simple vegetable make so many people cry?

② The secret is hidden inside the onion's cells. Onions take in **sulfur** from the soil while they grow. This sulfur is stored as a kind of chemical inside the tiny rooms of the onion, called cells. As long as the onion is whole, these chemicals stay safely inside. The onion looks quiet and harmless on the kitchen table.

③ But the moment your knife cuts through it, things change quickly. The cells break, and two groups of chemicals that are usually kept apart suddenly meet. They react together and form a new gas. This gas rises from the cut surface and floats into the air around your face. Scientists call it a kind of "tear-making" gas, or more correctly, a **chemical irritant**.

④ When this gas touches the thin wet layer on the front of your eyes, it changes into a mild acid. Your eyes do not like acid, so they send a quick message to the brain: "Wash it away!" The brain answers by asking your **tear glands** to produce more water. That is why tears come, even when you feel fine inside. It is simply the body protecting the eyes, just like it does when dust or smoke gets in.

⑤ Why have onions developed such a strange weapon? Scientists believe it is a kind of self-defence. In the wild, the gas drives away small animals that want to eat the onion roots. The onion that cries the most animals away has the best chance to survive and have more baby onions. So the tears in your kitchen are really part of a very old battle between plants and hungry animals.

⑥ If you do not want to cry while cooking, there are a few easy tricks. Put the onion in the fridge for 10 minutes before cutting — cold slows the reaction. Use a **sharp** knife, which breaks fewer cells than a blunt one. Cut near an open window, or even wear swimming glasses for fun. Next time tears roll down while you help your parents cook, smile a little: you are taking part in an ancient story of plant chemistry.

根据短文内容，选择最佳答案。

1. What does an onion store from the soil while it grows?

A. Salt.
B. Sulfur.
C. Sugar.
D. Iron.

2. What happens in an onion the moment a knife cuts it?

A. Nothing changes inside.
B. The cells break and two chemicals meet to form a gas.
C. The onion turns into water.
D. The onion becomes sweeter.

3. Why do our eyes produce tears when this gas arrives?

A. Because we feel sad.
B. Because the body wants to wash the mild acid away.
C. Because the gas is very cold.
D. Because our brain wants to rest.

4. What does the underlined word "irritant" in paragraph ③ most probably mean?

A. A sweet smell.
B. A soft sound.
C. Something that makes the body uncomfortable.
D. A kind of fruit juice.

5. Which is the BEST title for this passage?

A. A Short History of Chinese Cooking.
B. How to Grow Onions at Home.
C. The Science Behind Onion Tears.
D. Dangerous Vegetables You Should Never Eat.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段第1句 "Onions take in **sulfur** from the soil while they grow."
【解析】原文明确指出洋葱从土壤中吸收硫。A 盐、C 糖、D 铁在文中都未提及，为无中生有。故选 B。

2.
【答案】B
【题型】细节理解题

【定位】第③段第2句 "The cells break, and two groups of chemicals ... meet. They react together and form a new gas."
【解析】切洋葱时细胞破裂，两种化学物质相遇产生新气体。A 与原文矛盾；C "变成水"、D "变甜"均属无中生有。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第④段第2–3句 "Your eyes do not like acid, so they send a quick message to the brain: 'Wash it away!'"
【解析】气体在眼表变为弱酸，身体靠眼泪冲洗掉。A "感到悲伤"与原文"nobody is sad"矛盾；C 气体冷、D 大脑休息都属无中生有。故选 B。

4.
【答案】C
【题型】词义猜测题
【定位】第③段末句以及第④段整体语境
【解析】"irritant" 是能让眼鼻不适、引起流泪的气体，推断其含义是"令人不适的东西"。A 甜味、B 柔声都与让人流泪的刺激作用相反；D 果汁与化学气体无关。故选 C。

5.
【答案】C
【题型】主旨大意题
【定位】全文主题句第①段末与各段解释
【解析】文章解释洋葱为何使人流泪的科学原理及防护小贴士，中心是"洋葱流泪背后的科学"。A 中餐历史、B 家庭种植、D 危险蔬菜均与文章中心不符。故选 C。

词汇注释（超纲词）

- **sulfur** [n.] 硫
- **irritant** [n.] 刺激物
- **tear gland** [n.] 泪腺
- **acid** [n.] 酸（此处指弱酸性物质）